

Miguel Cortes Munoz

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SOFTWARE DEVELOPER

I'm an entry-level software engineer who enjoys building things end to end, from a full-stack web app with a responsive UI and optimized SQL schema to a custom network protocol built from raw sockets. I pick up new tools fast and work well with teams to turn ideas into something that actually runs. Right now I'm looking for a role where I can keep growing as a developer and take on real technical challenges.

TECHNICAL SKILLS

Languages: C++, Python, SQL, PHP, HTML

Frameworks & Libraries: ReactJS

Systems & Networking: Linux (Ubuntu), TCP/UDP protocols, sliding window protocol, CRC error detection, cryptography, IIoT Systems, Real-time Data Processing

Databases & Web: SQL schema design, normalization (up to 3NF), full-stack web development (HTML, PHP), REST concepts

Core CS: Data structures & algorithms, computer architecture, object-oriented programming, CI/CD, Agile, Scrum and unit testing, DevOps Practices

Tools: Git, GitHub, Docker

PROJECTS

UDP-to-TCP File Transfer Protocol

- Designed and implemented a reliable file transfer system over LAN using C++ and raw UDP sockets, replicating TCP-like delivery to support real-time data processing in constrained networks
- Engineered CRC error checking and a sliding window protocol, applying data structures and algorithms to guarantee data integrity and ordered packet delivery
- Applied object-oriented programming principles to modularize protocol components and facilitate maintainability

Car Dealership Database System

- Designed and normalized a relational database schema (3NF) using SQL to efficiently manage high-volume inventory, customer, and transaction data
- Built and deployed a full-stack web interface with PHP and HTML, leveraging RESTful APIs to support live SQL queries and CRUD operations, improving data accessibility

Predicting Steam Game Reception

- Engineered 507 features from a 40,000+ game dataset, including one-hot encoded genre/tag vectors and a Wilson Score target variable, to train a binary classifier
- Achieved 0.73 ROC-AUC; applied chi-square significance testing and exploratory data analysis to surface actionable patterns for indie developers

SVHN Digit Classification

- Trained and benchmarked machine learning classifiers (Gaussian Naive Bayes vs. Logistic Regression) on the Street View House Numbers dataset for real-world digit recognition
- Evaluated and documented accuracy trade-offs between classification approaches, demonstrating model selection and performance analysis skills

P-Center Hospital Placement Optimizer

- Solved a P-Center optimization problem to determine ideal hospital locations across a region, minimizing the maximum distance from any point to a facility
- Applied graph algorithms and combinatorial problem-solving to a real-world resource allocation scenario, integrating Statistics Canada data for geographic accuracy

EXPERIENCE

Scuba SEAsons

Jun 2026 - Present

Frontend Developer

Calgary

- Contribute frontend features and improvements to an established production codebase, working within existing React components, application architecture, and development conventions
- Implement UI changes and fix frontend issues while maintaining consistency with existing functionality and design
- Collaborate with the project owner/development team through Git-based workflows, code reviews, and issue tracking
- Navigate and modify an unfamiliar codebase, debugging existing components and integrating changes without disrupting established functionality

Arrow Workforce Solutions

Jun 2026 - Present

Programmer Analyst

- Configured and deployed online surveys in LimeSurvey based on client requirements and detailed questionnaire specifications
- Built survey logic using conditional statements, branching workflows, and validation rules to keep data collection accurate and user flow smooth
- Ran functional testing and troubleshooting to catch and resolve configuration issues before deployment
- Worked closely with the team to refine requirements and make sure survey designs were implemented correctly while keeping data integrity intact

Respite

Sep 2025 - Jun 2026

Respite Private Tutor

- Delivered one on one instruction, translating complex concepts into clear explanations for a non technical learner
- Tracked progress over time and adjusted teaching approach based on what was and wasn't working
- Diagnosed knowledge gaps and built personalized learning strategies tailored to the student's pace and needs

Modern Rentals

May 2022 - Aug 2024

Event Tenter

Coordinated a team of staff to streamline workflow and boost overall productivity

- Consistently hit assigned objectives within tight deadlines
- Delegated tasks and gave constructive feedback to help team members perform at their best
- Onboarded and trained new staff on operational procedures and inventory systems, helping them get up to speed quickly

EDUCATION

Bachelor of Science, Computer Science

Sep 2021 - May 2026

University of Lethbridge

Lethbridge, AB

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Artificial Intelligence, Software Engineering, Database Design, Discrete Structures